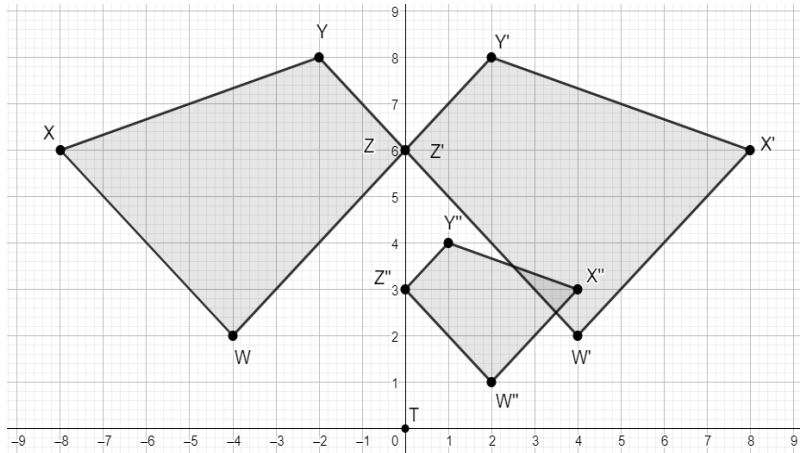


A sequence of transformations of WXYZ is shown on the coordinate grid. WXYZ is reflected then dilated, where W'X'Y'Z' is the reflection and W''X''Y''Z'' is the dilation of W'X'Y'Z'.



Complete the descriptions:

- 1. WXYZ was reflected over the y-axis to create W'X'Y'Z'.
- 2. W'X'Y'Z' was dilated using a scale factor of 1/2 centered at T to create W''X''Y''Z''.
- 3. Complete the table:

Transformation	Algebraic Description
Reflection	$(x,y) \rightarrow (-x,y)$
Dilation	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$

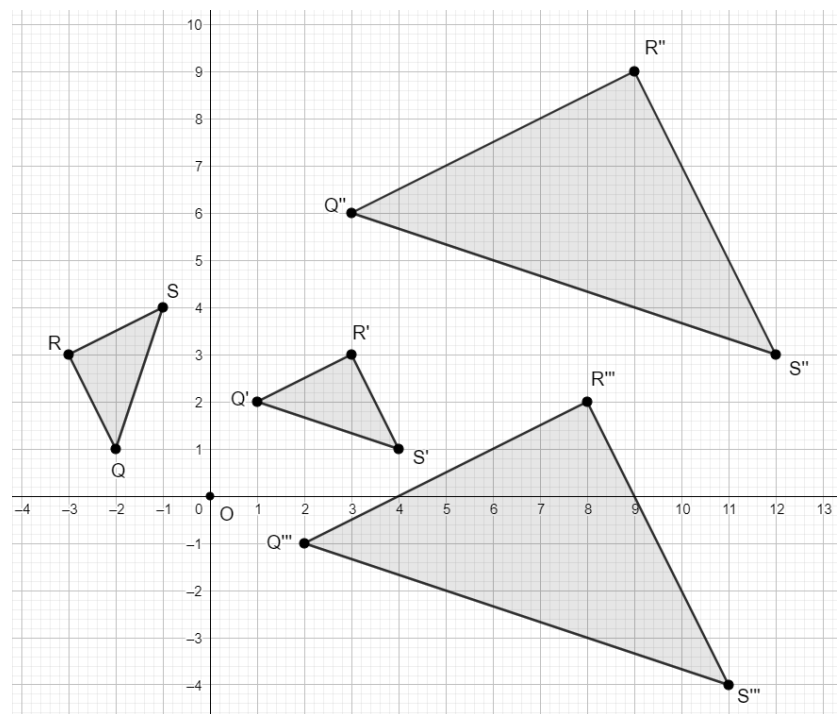
- 4. Determine the coordinates if WXYZ is first dilated then reflected:

Preimage WXYZ		Dilation W'X'Y'Z'		Reflection W''X''Y''Z''	
W	(-4,2)	W'	(-2,1)	W''	(2,1)
X	(-8,6)	X'	(-4,3)	X''	(4,3)
Y	(-2,8)	Y'	(-1,4)	Y''	(1,4)
Z	(0,6)	Z'	(0,3)	Z''	(0,3)

- 5. Based on the results above, the sequence of transformations commutative.

✗ is
o is not

$QRS, Q'R'S', Q''R''S'',$ and $Q'''R'''S'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- QRS is first

☐ translated

 to $Q'R'S'$, then

☐ translated

 to $Q''R''S''$,

☐ reflected

☒ rotated

☐ dilated

☐ reflected

☐ rotated

☒ dilated
- and finally

☒ translated

 to $Q'''R'''S'''$.

☐ reflected

☐ rotated

☐ dilated
7. Describe the transformation for $QRS \rightarrow Q'R'S'$.
Rotation 90° clockwise about the origin
8. Describe the transformation for $Q'R'S' \rightarrow Q''R''S''$.
Dilation by a scale factor of 3 from the origin
9. Describe the transformation for $Q''R''S'' \rightarrow Q'''R'''S'''$.
Translation 1 unit left and 7 units down
10. Write the algebraic description for the sequence of transformations from 7-9:

Transformation	Algebraic Description
Rotation	$(x,y) \rightarrow (y,-x)$
Dilation	$(x,y) \rightarrow (3x,3y)$
Translation	$(x,y) \rightarrow (x-1,y-7)$